INCH-POUND
MIL-DTL-24643/29D
22 August 2002
SUPERSEDING
MIL-C-24643/29C
14 March 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSMU
(INCLUDING VARIATION LSMUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

This cable can be manufactured in three variations: LSMU and LSMUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 20-7.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation white. Minimum average wall thickness 0.013 inch. Standard identification code applied by method 1.
Fourth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, will-rounded assembly. Cabling sequence shall be consecutive starting with no. 1 from center outward.
Fifth - Binder tape, applied helically with overlap.

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Construction variant type LSMU

Sixth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.030 inch.

Construction variant type LSMUS

Sixth - Braided shield of tin coated copper.
Seventh- Spirally wrapped 0.002 inch thick polyethylene (polyester) separator tape with 25 percent minimum overlap.
Eighth - Braided shield of tin coated copper.
Ninth - Binder tape applied helically with overlap.
Tenth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.030 inch.

TABLE I. Details.

Military[1/] part no. M24643/29	Size [2/]	Number of conductors	Diameter over shielded core	Overall diameter (inches)			
				TYPE LSMU		TYPE LSMUS	
			(inch nom.)	min	max	min	max
-01UN	14	14	0.340	0.365	0.400	0.425	0.460

1/ When double shield is required, see MIL-DTL-24643 for configuration of part number.
2/ Type LSMU and LSMUS.

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor resistance - ohms/1000 feet at 25°C, maximum	10.93
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Shield to shield (type LSMUS only, overall shields only	200
Insulation resistance - megohms -1000 feet, minimum	
Conductor to conductor	2000
Shield to shield (type LSMUS only)	100
Jacket flaws (LSMUS only)	No failure
Conductor and shield (LSMUS only) continuity	No failure

Group A:	
Visual and dimensional	No failure

Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (when tested at 200°C)	50
Cold bending, cable - 6 inch maximum mandrel diameter	No damage

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Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 700
Elongation - percent, minimum 250
Cross-linked polyethylene
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals - (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged, minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (conductor) - cycles minimum 25
Permanence of printing (jacket) - cycles, minimum 125

Shield - conformance to material,
construction and coverage ..... No failure
Surface transfer impedance (type LSMUS only)
milliohms per meter, maximum 700
EMP response time - dB, minimum 60

Group D:
Flame propagation (cable) ..... No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH:

2500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-027)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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